

ALEXANDRE MIGUEL SILVA

Computer Engineering Finalist • SoftwareDevelopment • APIs, Automation & Integration • FullStack, FrontEnd & BackEnd

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PROFILE

Computer Engineering finalist with hands-on development experience across Python, Java, JavaScript/TypeScript, C and C#, built through team projects and self-directed work. Comfortable across the full stack — REST API design and integration, relational data modelling and front-end development — with a particular focus on automation and connecting systems. Recently built a full-stack incident management platform (FastAPI, PostgreSQL, React) covering alert triage, prioritisation, asset and vulnerability tracking and workflow automation. Four years of client-facing experience in technology retail alongside a full-time degree. Seeking a first professional role in software engineering, integration or technology consulting, with a long-term interest in Security Operations.

PROFESSIONAL EXPERIENCE

Worten Portugal Porto, Portugal
Sales Assistant (Part-time - Alongside full-time Degree) 2022 – Present

- Advise customers across consumer electronics, telecommunications and IT, **translating technical concepts for non-technical audiences in a high-volume retail environment.**
- Diagnose and resolve product and configuration issues, manage after-sales cases and deliver against individual and store commercial targets.
- Four years of sustained client-facing work while completing a full-time Computer Engineering degree.

KEY PROJECTS

Incident Management Platform — *Python/FastAPI, PostgreSQL, React* — *Personal project* 2026 – Present

- Full-stack platform managing the complete incident lifecycle: **ingestion, classification, severity-based prioritisation, asset registry, vulnerability tracking, state-driven workflows and audit trail.**
- Designed the relational data model and a **REST API integration layer**, applying ITIL incident-management principles to the workflow design.
- End-to-end ownership: FastAPI backend, PostgreSQL database, React front end.

GTM Lead Agent — *Python, REST APIs, LLM Integration* — *Solo project* June 2026 - July 2026

- End-to-end Python pipeline that takes a raw list of leads, enriches each company through a public API, and uses that context to generate a personalised outreach email — returning the full enriched dataset as a new CSV.
- Applies LLM integration to consolidate scattered company information into usable sales context, automating a task that is otherwise fully manual. Covered by an automated test suite.

Karting Management System — *Java, Spring Boot, Desktop + Web* — *Team project* May 2026 - June 2026

- Management system for a karting circuit covering track registration and configuration, race scheduling, customer records and day-to-day operations.
- Implemented booking and scheduling logic with conflict detection, preventing overlapping races and resource clashes across tracks.
- Built to a commercial standard as a product deployable to a real karting operator, with modular ownership, Git version control and agile practices.

License Management System — *C, CLI, Data Structures* — *Solo project* Mar 2026 - April 2026

- Software licence management system with authentication and multi-level access control, full CRUD over licences, renewal tracking and report generation.
- Data persistence implemented from scratch using linked lists and binary files, with input validation enforcing role separation and data integrity at application level.

2D Games — *JavaScript, Phaser 3, C#, Unity* — *Team project* April 2025 - May 2025

- Academic team projects covering game loops, physics, collision handling, level design and collaborative development.

EDUCATION

Polytechnic University of Viana do Castelo (UNIPVC) Sep 2022 – Jun 2027

BSc in Computer Engineering

- Programming and Data structures, Databases, Computer networks, Operating systems, Software engineering and Distributed systems.

CERTIFICATIONS & CONTINUOUS LEARNING

Certifications in progress: ISC2 Certified in Cybersecurity (CC) • ITIL 4 Foundation (Expected conclusion in February 2027)

Security fundamentals: NIST Cybersecurity Framework • OWASP Top 10 • MITRE ATT&CK

Hands-on practice: TryHackMe Learning paths • Controlled home lab covering penetration testing fundamentals, including self-built database-driven web applications used as targets to test and validate acquired knowledge.

Self-directed learning: Enterprise service management and incident workflow platforms.

SKILLS, TOOLS, METHODOLOGIES & LANGUAGES

Programming: Python • JavaScript • TypeScript • Java • C# • C • SQL

Web & Frameworks: React • Node.js • FastAPI • Spring Boot • HTML • CSS • Flutter • Angular

Databases: PostgreSQL • MySQL — Data Modelling, Queries and Administration

Tools & Methodologies: Git • GitHub • Docker • VS Code • IntelliJ IDEAPgAdmin • Claude • ChatGPT • Trello • Microsoft365 • Agile (Scrum)

Integration & Automation: REST APIs • LLM / AI integration • Scripting and process automation • Automated testing

Security & Operations: Incident lifecycle management • Alert classification and Prioritisation • Asset and vulnerability tracking • Audit trails • Role-based access separation • ITIL-aligned incident handling

Languages: Portuguese (Native) • English (B2)